RL Circuit Transient Response

1. Definition

The transient response of an RL circuit is the temporary change in current and voltage when a switch is turned ON or OFF. It occurs because an inductor opposes sudden changes in current. The transient period ends when the circuit reaches a steady state.

2. RL Circuit

An RL circuit consists of a resistor (R) and an inductor (L) connected in series with a DC voltage source. The resistor limits the current, while the inductor stores energy in its magnetic field.

3. Transient Response During Switching ON

When the switch is turned ON, the current does not reach its maximum value immediately. The inductor produces a back EMF that opposes the increase in current. Therefore, the current increases gradually and exponentially until it reaches its steady-state value.

Equation:

$i(t) = (V/R)(1 - e^{-t}/\tau)$

Where:

- V = Supply voltage
- R = Resistance
- L = Inductance
- τ = Time constant
- V/R = Final steady-state current

4. Transient Response During Switching OFF

When the supply is disconnected, the inductor releases its stored magnetic energy. The current decreases gradually and exponentially from its initial value to zero. The inductor generates a voltage to oppose the decrease in current.

Equation:

$i(t) = I_o e^{-t}/\tau$

Where:

- I_o = Initial current
- t = Time elapsed
- τ = Time constant

5. Time Constant of an RL Circuit

The time constant is the time required for the current to reach 63.2% of its final value during growth, or decrease to 36.8% of its initial value during decay.

Formula:

$\tau = L/R$ seconds

Where:

- L = Inductance in henry (H)
- R = Resistance in ohms (Ω)
- τ = Time constant in seconds

6. Important Characteristics

1. An inductor opposes sudden changes in current.

2. At the instant of switching ON, the current is zero if the inductor is initially unenergized.
3. During switching ON, current increases exponentially.
4. During switching OFF, current decreases exponentially.
5. In steady-state DC conditions, an ideal inductor behaves like a short circuit.
6. The transient response is practically complete after five time constants (5τ).

7. Time Constant Table

Time	Current Growth	Current Decay
τ	63.2%	36.8%
2τ	86.5%	13.5%
3τ	95%	5%
4τ	98.2%	1.8%
5τ	99.3%	0.7%

8. Conclusion

The transient response of an RL circuit describes how current changes with time when the circuit is switched ON or OFF. The rate of change depends on the time constant, $\tau = L/R$. A higher inductance increases the time required to reach steady state, whereas a higher resistance decreases it.